

NIST Face Morph Detection Evaluation (FATE MORPH)

Mei Ngan

National Institute of Standards and Technology (NIST)
U.S. Department of Commerce

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NIST's morph detection benchmark

F RTE

FACE RECOGNITION
TECHNOLOGY EVALUATION

RECOGNITION: **WHO** IS IN AN IMAGE

1:1 VERIFICATION

SAME PERSON OR NOT?

1:N SEARCH

WHO? WHERE? WHEN?

FACE IN VIDEO 2024

1:N ON NON-COOP PEOPLE

TWINS DISAMBIGUATION

SAME PERSON, OR TWIN?

FATE MORPH

- 2018 - present
- S-MAD
- D-MAD
- FR vulnerability
- Image reversal

FATE

FACE ANALYSIS
TECHNOLOGY EVALUATION

ANALYSIS: **ABOUT** IN AN IMAGE

MORPH DETECTION

TWO PEOPLE IN ONE FACE?

QUALITY + DIAGNOSTICS

HOW BAD IS THIS PHOTO

PAD

SUBVERSIVE PHOTO?

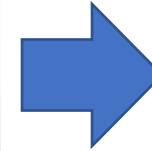
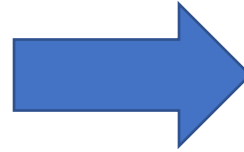
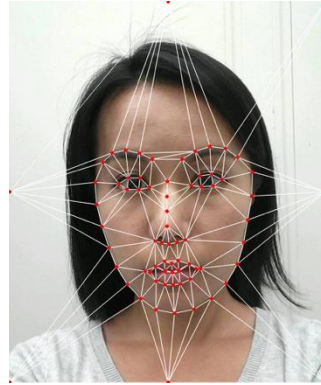
AGE ESTIMATION

HOW OLD? OLD ENOUGH?

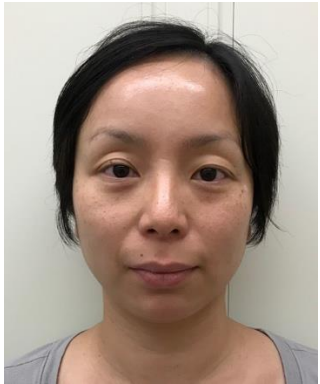
Face morphing



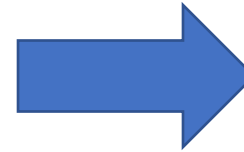
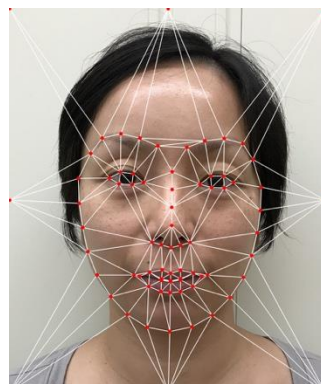
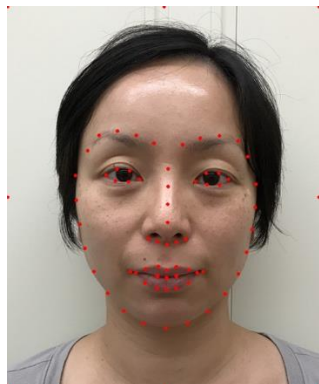
Subject A



Morph (A + B)



Subject B



Face morphing generates an image that visually resembles both contributing subjects

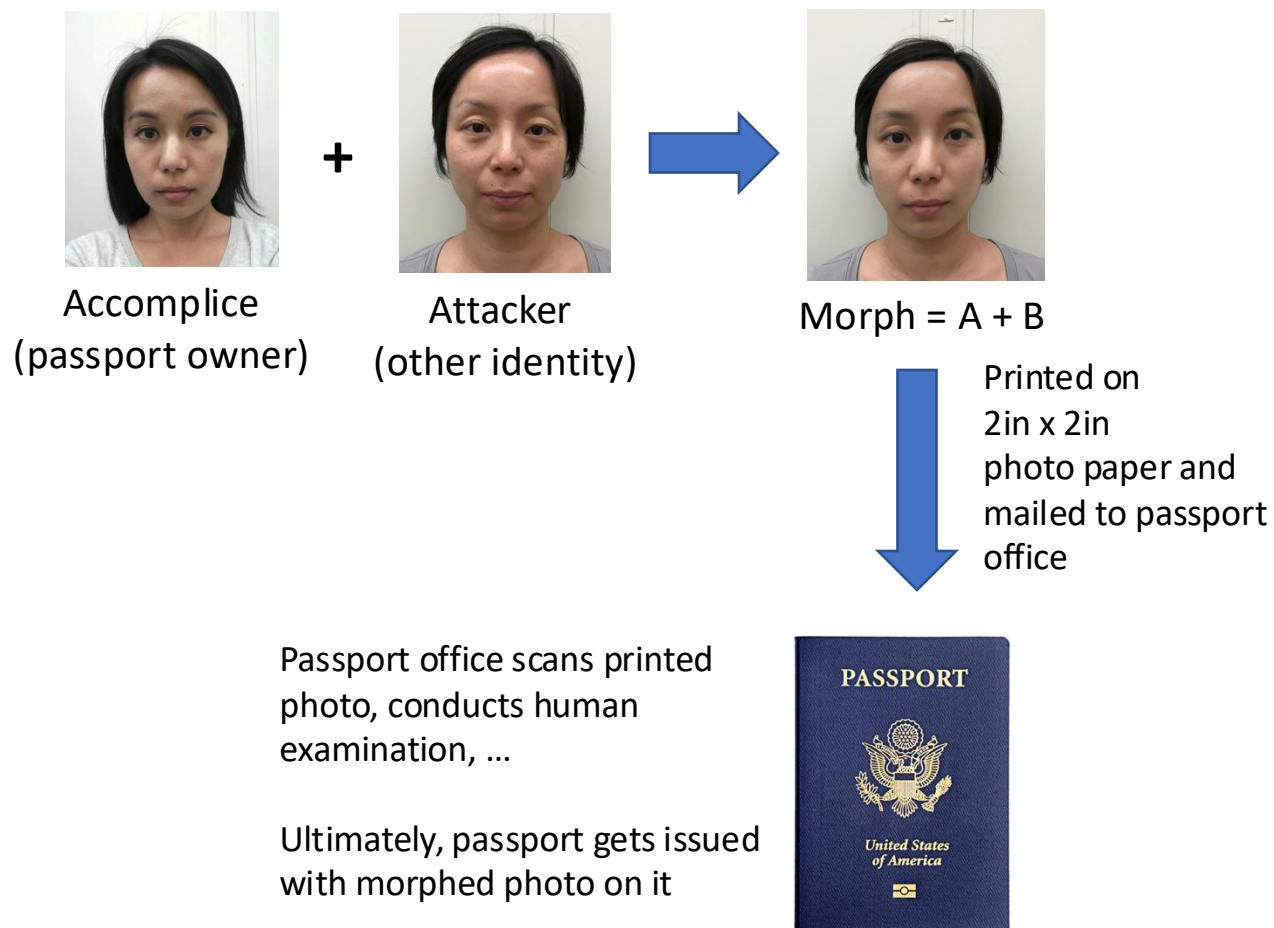
Problem: face recognition matches **both** persons



- Both people can authenticate against a morph
- All modern face recognition algorithms tested by NIST and operational matchers tested by parts of the U.S. Government are vulnerable to morphs

Threat: one document, **two** users

PASSPORT APPLICATION



Some U.S. passport applications are susceptible to manipulation of user-submitted photos.

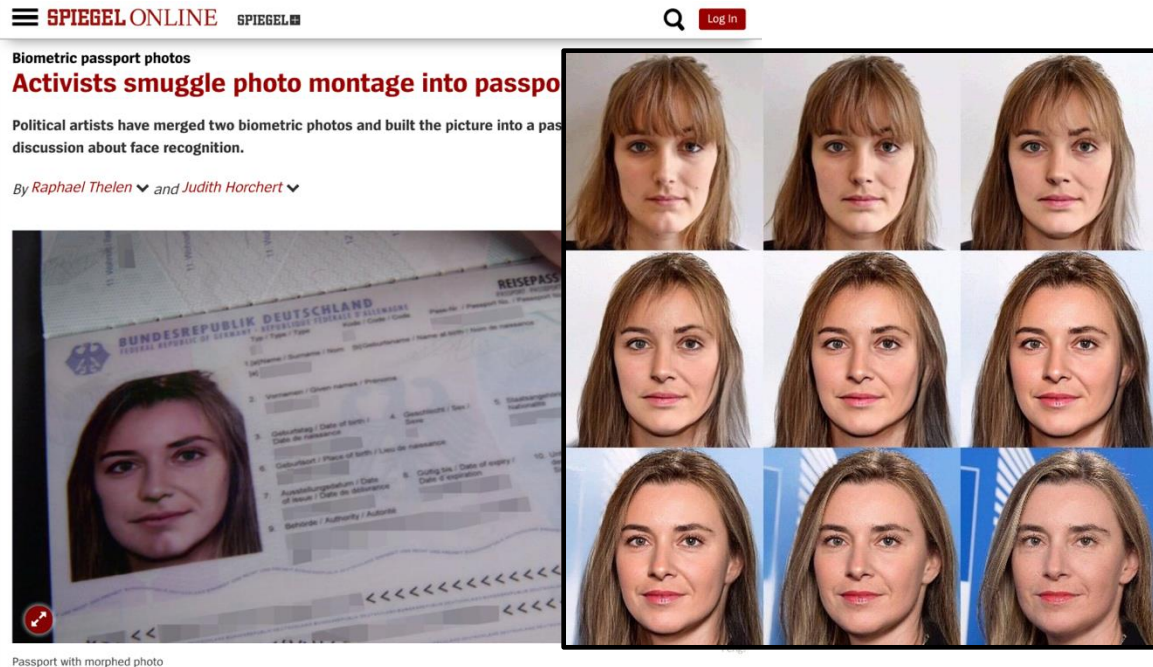
Other countries also accept user-submitted photos for identity credential applications.

Risk #1: Organizations that accept user-submitted photos for enrollment are at risk of morphing attack.

Risk #2: Organizations that may have to process ID documents from other countries that are vulnerable to morphs may also be at risk.

Real cases of morphing

NIST



Sept. 22, 2018: Member of German activist group successfully applies for a passport with a morphed image (containing Federica Mogherini, High Representative of the Union for Foreign Affairs and Security Policy)

Source (9/22/2018): <http://www.spiegel.de/netzwelt/netzpolitik/biometrie-im-reisepass-peng-kollektiv-schmuggelt-fotomontage-in-ausweis-a-1229418.html> via Google Translate

OVER 1000+ MORPHING CASES REPORTED ACROSS THE EU

Source: Presentation by Christoph Busch, Professor at NTNU/Hochschule Darmstadt at the International Face Performance Conference (IFPC) 2020, October 30, 2020
<https://www.nist.gov/itl/iad/ig/ifpc-2020-conference-presentations-and-videos>

SINCE 2020, OVER 40 MORPHING CASES WERE DETECTED IN SLOVENIA

Source: Presentation by Matjaz Torkar, Deputy Commander of Station, Airport Police Station Brnik Slovenia at the International Face Performance Conference (IFPC) 2022, November 17, 2022
<https://www.nist.gov/itl/iad/ig/ifpc-2022-conference-presentations-and-videos>

NIST FATE MORPH Evaluation [Ongoing]



Automated Face Morph Detection Evaluation

- Independent, **sequestered** evaluation of morph detection capabilities across diverse datasets
- “Black-box” testing
- Ongoing testing + public reporting (report + interactive webpage)

Use Cases

- Single-image morph detection (S-MAD)
- Two-image differential morph detection (D-MAD)
- 1:1 morph acceptance (FR resistance against morphing)
- Demorphing (Image reversal)

Recent participants

- **Academic:** Norwegian University of Science and Technology (NO), Fraunhofer Institute for Telecommunications Heinrich Hertz Institute (DE), Universidade de Coimbra (PT), West Virginia University (US), University of Bologna (IT), Hochschule Darmstadt
- **Commercial:** Idemia (FR), Neurotechnology (LT), Vision-box (PT), Secunet (DE), Kempelen Institute of Intelligent Technologies (SK)



FATE MORPH report and leaderboard last updated February 2025
Ongoing morph detection submissions accepted! Google: FATE MORPH

FATE MORPH use case: morph detection

Single-
image
(S-MAD)

Morph detection with single
image in isolation
(e.g., document issuance)

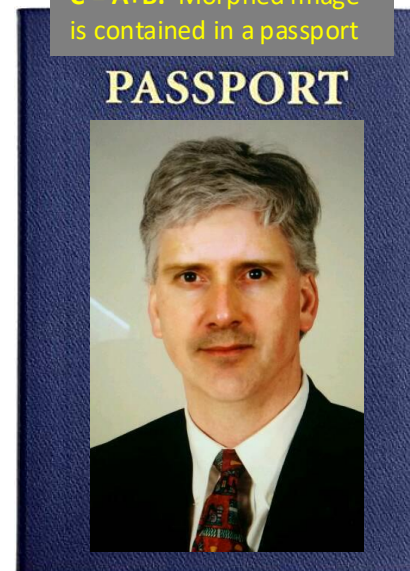


DOCUMENT ISSUANCE:
Suspect image in isolation

Differential
(D-MAD)

Morph detection with
additional live capture image
(e.g., border crossing)

$C = A+B$. Morphed image
is contained in a passport



A. Images of this image
not available during
authentication



B2: This image represents a live
capture during an eGate border
crossing, say.



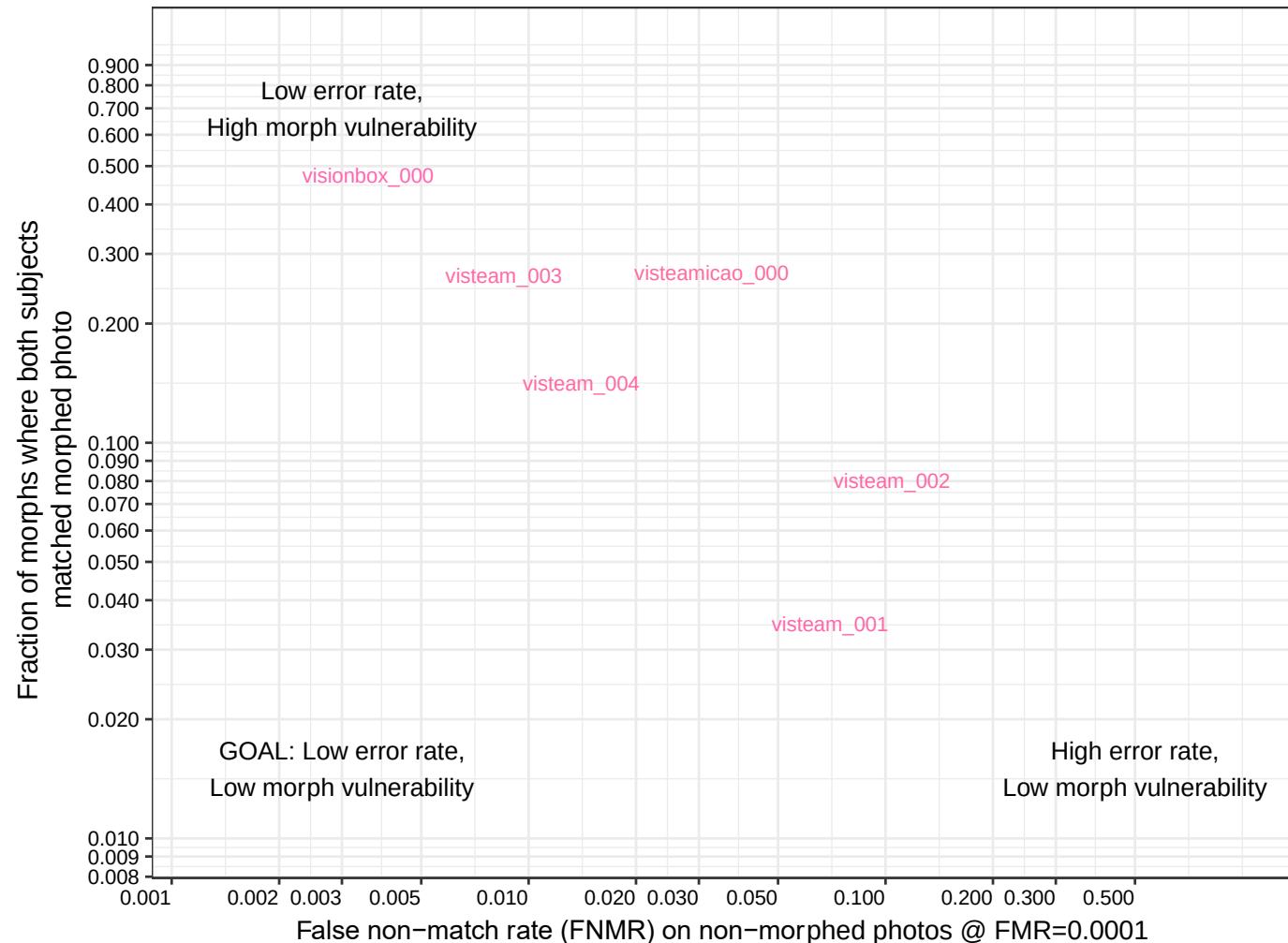
BORDER CROSSING: Suspect image + live image

FATE MORPH use case: morph resistant 1:1 face recognition



Source: NIST + USG employee with permission

**FR algorithm resistance
against morphing: Morph +
bona fide of contributing
subjects**

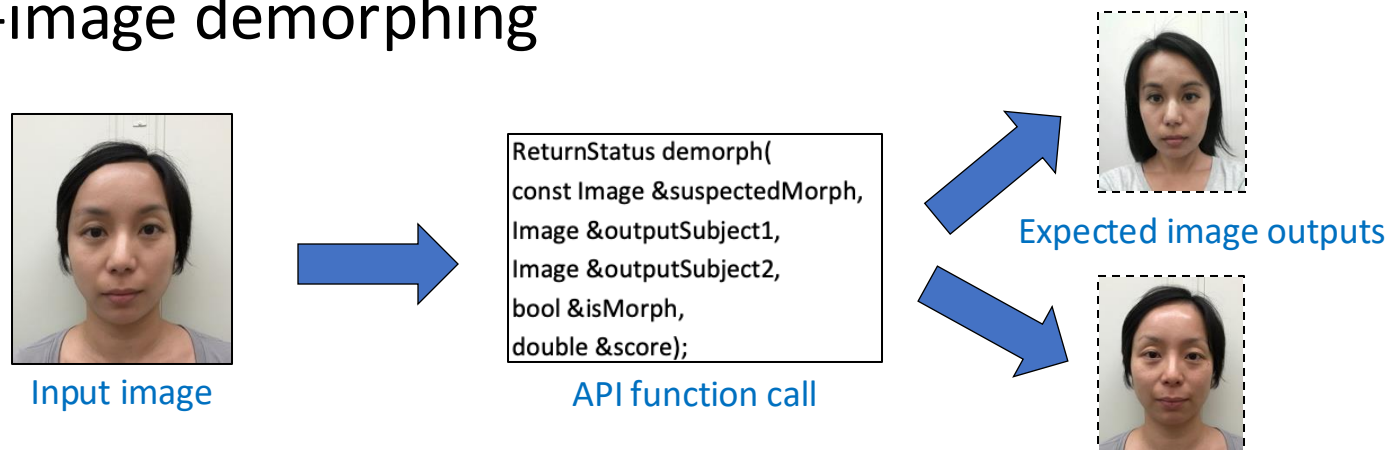


Track
FRTE 1:1
FATE MORPH

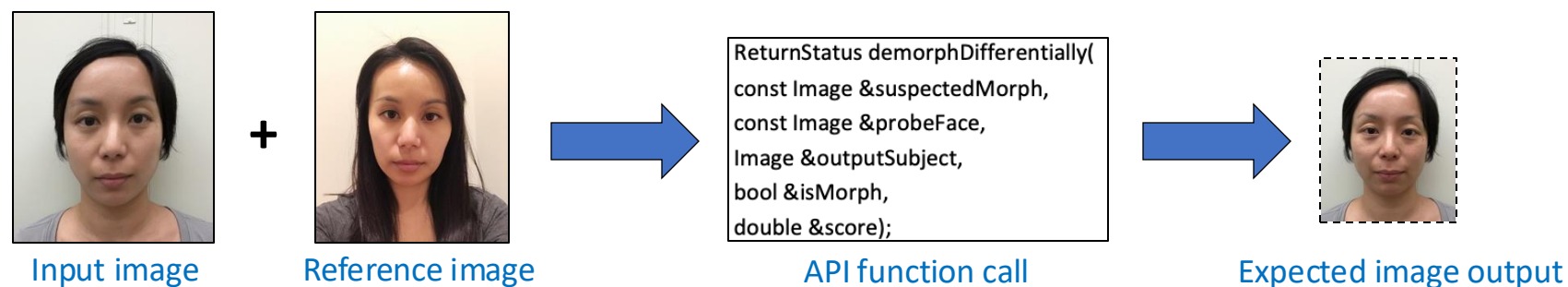
FATE MORPH use case: demorphing

Demorphing is the process of recovering the original faces/identities used to generate a morph

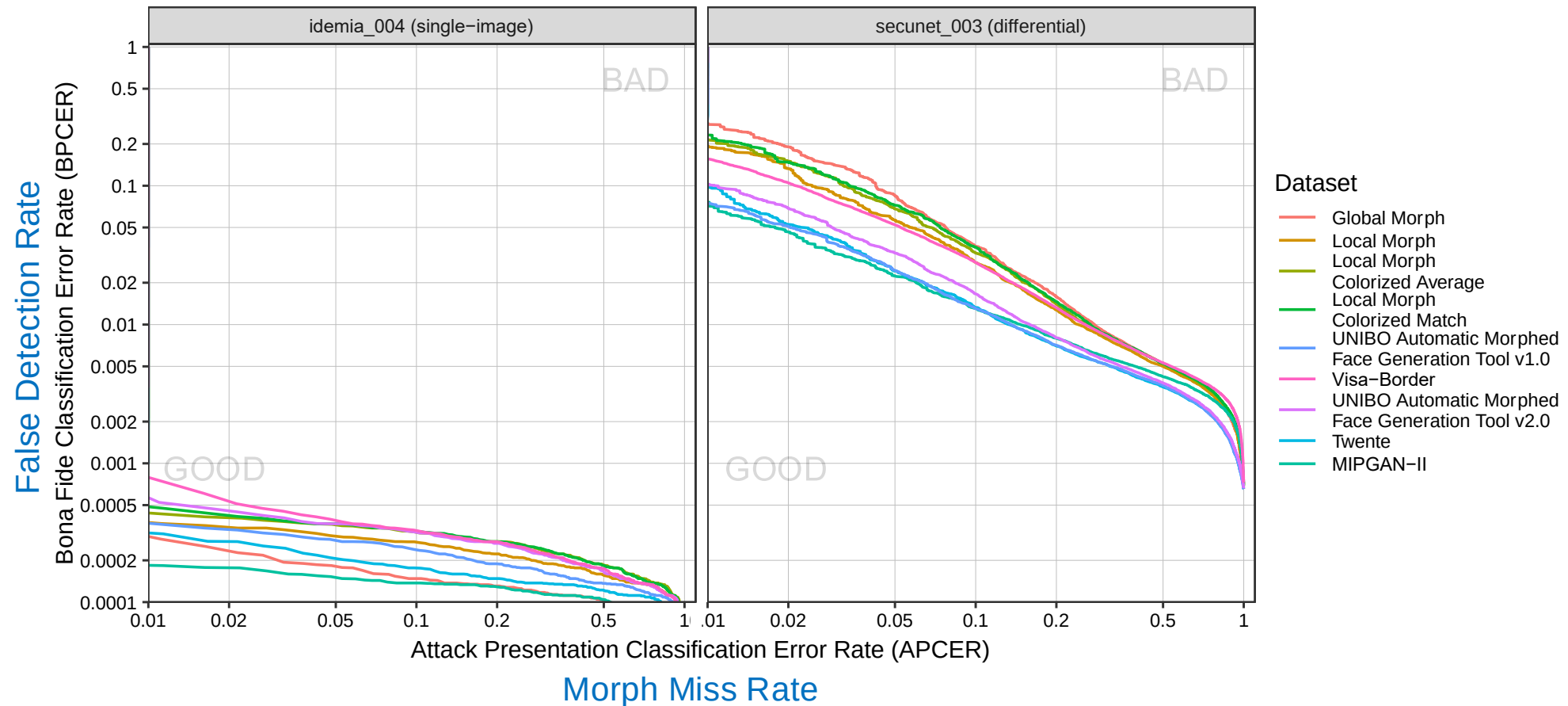
- Single-image demorphing



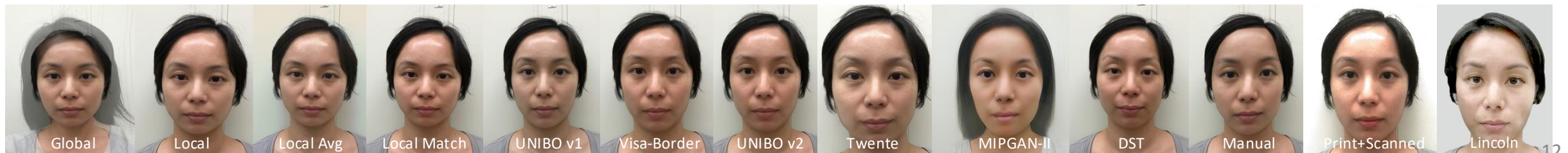
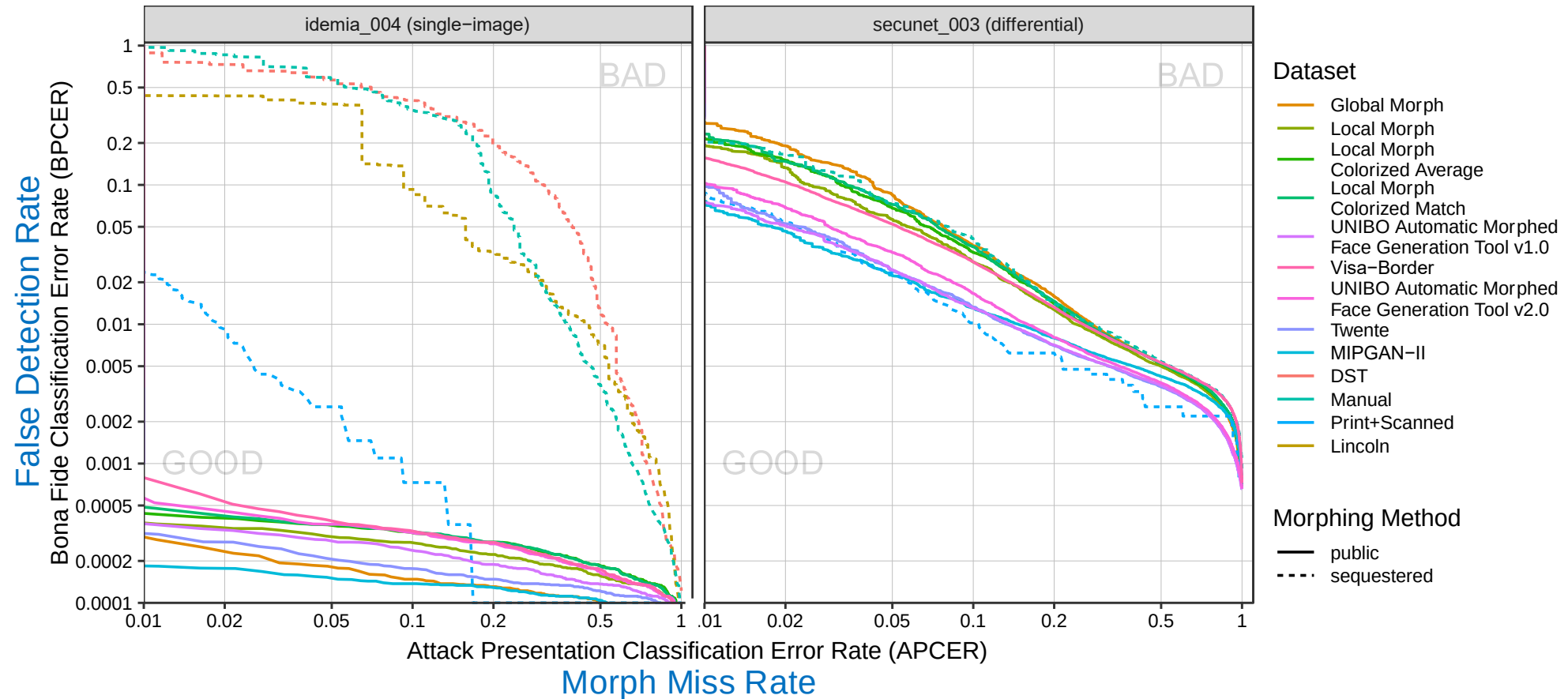
- Differential demorphing



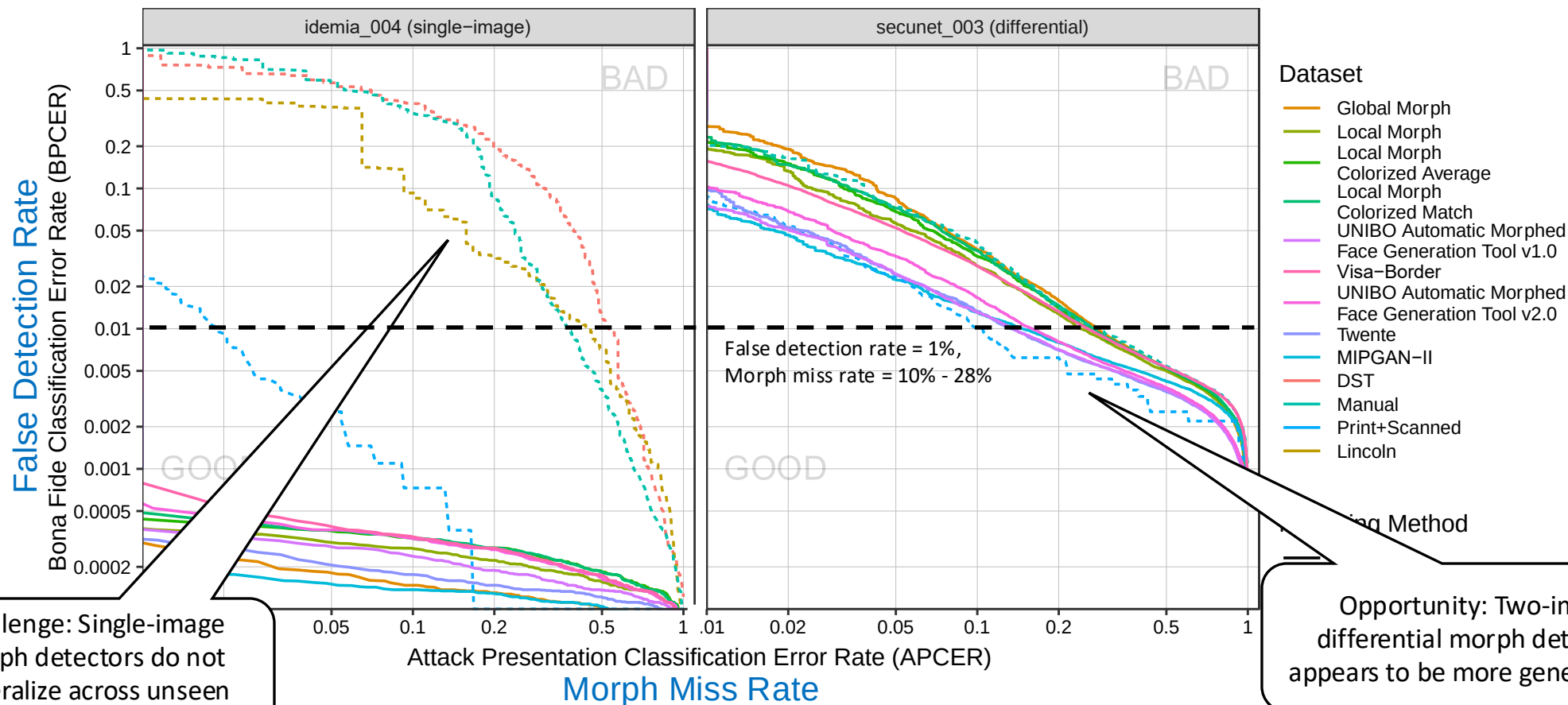
Automated morph detection (2025): morphing methods from public domain

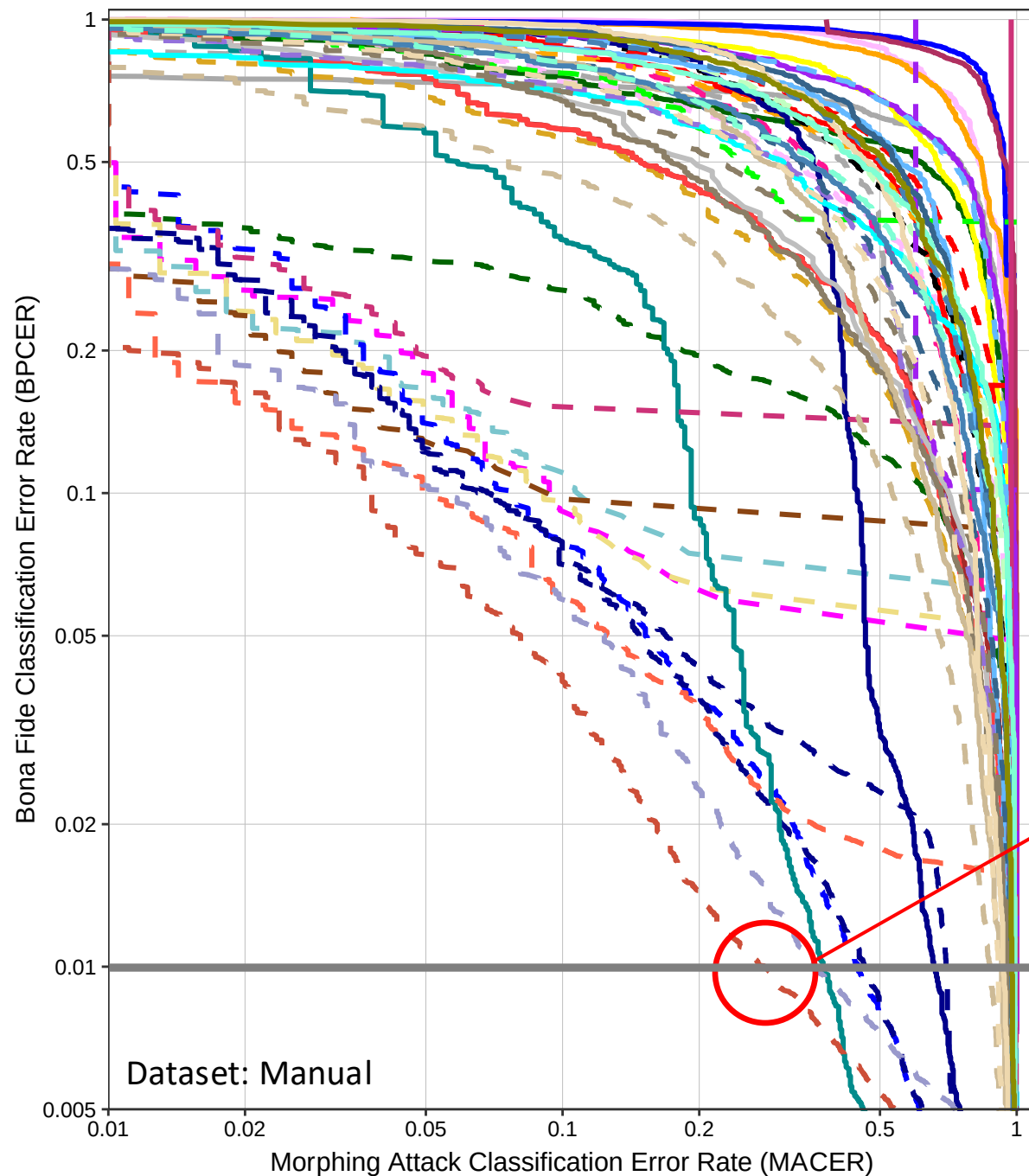


Automated morph detection (2025): high quality/sequestered morphing methods



Automated morph detection (2025): high quality/sequestered morphing methods





Morph detection capability is close to valuable:
Run silent pilots

False Accusation Rate, BPCER = 0.01

Proportion of morphs **not** detected,
MACER = 0.28

— single-image (S-MAD)

- - - differential (D-MAD)

Morph attack detection in operations

DIFFERENTIAL MORPH DETECTION IN BORDER CONTROL (D-MAD)

- Suspected morph from passport
- Live border crossing photo → not a morph

FATE MORPH RESULT

- D-MAD technology built on FR quantifies unusual face difference
- S-MAD technology potentially trained on public domain morphing methods may have value
- Ready for operational “behind-the-scenes” pilot configured with sufficiently low “false accusation rate” (BPCER = 0.01 or 0.001, say)

DOCUMENT SCOPE

- Algorithm selection
- Deployment: ID application, inbound immigration, forensic investigation
- Algorithm configuration – low BPCER
- Human-led investigative process after a candidate morph detected



NIST Interagency Report
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Face Analysis Technology Evaluation (FATE)

4B: Guidelines for Operational Use of Morph Detectors

Draft Stage

Mei Ngan
Patrick Grother
Information Access Division
Information Technology Laboratory

This publication is available free of charge from:
<https://doi.org/10.6028/NIST.XXX.XXXX>

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U.S. Department of Commerce
Howard Lutnick, Secretary

National Institute of Standards and Technology
Craig Burkhardt, Acting Under Secretary of Commerce for Standards and Technology and Acting NIST Director

**MORPHING:
THE SOLUTION!!!**

TRUSTED CAPTURE!

DENY ALL OPPORTUNITIES TO INTRODUCE A MORPH

WHEN

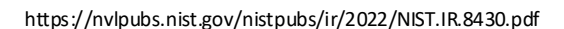
- ISSUANCE + RENEWAL

HOW

- TRUSTED STAFF: LIVE ATTENDED CAPTURE
- TRUSTED HARDWARE:
 - TAMPER-PROOF BOOTH/DEVICE
 - DIRECT UPLINK TO RELYING PARTY
 - LIVENESS DETECTION

NLST

Expected result:	Strong hit, if previously encountered, OR No hit if not.
But for morphed image:	Possible weaker hit.



Use of 1:N face recognition for morph detection



Example Scenario: Passport Renewal



Bona fide search image
(mated – subject in database)

	Score	Rank	
	4.087	1	VERY HIGH SCORES @ RANK 1 + 2
	4.001	2	
	3.991	3	
	0.707	4	
	0.330	5	
	0.198	6	
	0.074	7	
	0.016	8	
...			



Morphed search image
(ONE subject in database)

	Score	Rank	
	3.142	1	(but reduced) HIGH SCORES @ RANK 1 + 2
	3.011	2	
	3.000	3	
	0.707	4	
	0.330	5	
	0.198	6	
	0.074	7	
	0.016	8	
...			



Morphed search image
(BOTH subjects in database)

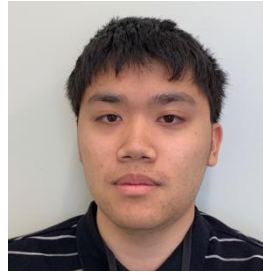
	Score	Rank	
	3.142	1	(but reduced) HIGH SCORES @ RANK 1 + 2
	3.110	2	
	3.028	3	
	3.011	4	
	3.000	5	
	0.198	6	
	0.074	7	
	0.016	8	
...			



Patrick
Grother



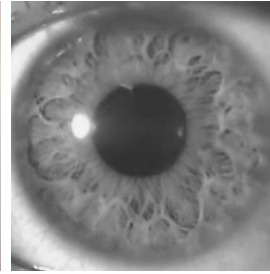
Kayee Hanaoka
(+ Mei Ngan)



Austin
Hom



(not)
Mei Ngan



Jim
Matey



Joyce
Yang



THANK YOU

MEI@NIST.GOV
FRVT@NIST.GOV

